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MSWDefTopic



Mixed Signal Workshop

Mixed Signal Workshop

Mixed Signal Workshop (MSW) is a tool for creating coherent clocking solutions used in testing mixed signal devices. The MSW generates solutions that consider DUT, DSP, and instrument-

specific constraints. The details of each clocking solution are stored as a timing set on a Mixed Signal Timing sheet in an IG-XL workbook. A timing set contains clocking and signal specific setup information for instruments, which can then be used in future tests.

The following sections describe what the MSW is, how it interacts with other IG-XL components, and some of the technical concepts behind the tool.

- [About MSW](#)
- [MSW Interaction with Other IG-XL Components](#)

MSW feature descriptions and usage instructions are provided in the following sections:

- [Parts of the MSW Window](#)
- [Using the MSW Tool](#)
- [MSW Solver Tool](#)

Related Topics

- [About WaveDesigner](#)
- [Mixed Signal Timing Sheet](#)





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Mixed Signal Workshop : About MSW

About MSW

The Mixed Signal Workshop (MSW) tool helps you build reusable mixed signal timing solutions that meet coherence requirements. Calculating a timing solution yourself is a complex and iterative process. The MSW automates the process, while respecting constraints related to device timing, signals, analysis, throughput, and tester resources.

The output of the MSW is a timing set that contains a signal’s structural characteristics and the details of the clocking solution that you created. Timing sets are stored on a Mixed Signal Timing sheet in the IG-XL workbook and become a reusable timing solution for future tests.

You can assign timing sets created with the MSW as part of a test instance. A timing set lets the test instance know how to program timing related parameters for the instruments in the test. The timing related instrument parameters are applied when using a test element for the instrument in a test procedure.

The MSW is useful in determining timing solutions because it is an accurate and automated way to calculate timing relationships, while respecting imposed constraints. Also, if multiple instruments are used such that they are related to one another in the same test, the MSW helps you find solutions that synchronize all the instruments.

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Mixed Signal Workshop : MSW Interaction with Other IG-XL Components

MSW Interaction with Other IG-XL Components

MSW is used to create and edit timing sets that take into consideration DUT, DSP, and instrument-specific constraints. To balance all of these constraints, the MSW interacts with other IG-XL components, such as the WaveDesigner, to get signal and test procedure information. The MSW also communicates the resulting timing set to the test instances.

The following sections describe how the MSW tool interacts with IG-XL components:

- [MSW and Mixed Signal Timing Sheets](#)

- [MSW and Test Instances](#)

- [MSW and the WaveDesigner](#)

MSW and Mixed Signal Timing Sheets

MSW is used to create and edit timing sets that contain clocking and signal-specific setup information for instruments. Timing sets are stored on a Mixed Signal Timing sheet in the IG-XL workbook. Having a timing set stored on a Mixed Signal Timing sheet makes the data available during test program execution and for use in future test procedures. A workbook can contain more than one Mixed Signal Timing sheet, but that requires a Job List sheet. If a Mixed Signal Timing sheet does not exist when you try to save a timing set, the sheet is created.

For more information, see:

- [Mixed Signal Timing Sheet](#)

- [Job List Sheet](#)

MSW and Test Instances

A [Test Instances sheet](#) includes a Mixed Signal Timing column where you can specify the timing set to be loaded for a test. You can only specify one timing set per test. If a test requires more than one clocking scheme, you must create more than one subset in the timing set and switch subsets as necessary during the test.

For more information, see:

- [Using MSW Subsets](#)

- [Test Instances Sheet](#)

MSW and the WaveDesigner

The WaveDesigner tool creates simple or compound wave definitions independent of timing and instrument-specific information. These wave definitions define the shape of a signal. The MSW uses wave definitions to create the samples for the signal.

When setting up the source instrument block, you must specify a wave definition to use. To edit a wave definition, click the Edit button in the source block and the wave definition opens up in the WaveDesigner tool window. See [Signal Timing Information](#).

The WaveDesigner allows you to specify the shape of a signal, but it supplies no information about its dimensions (such as its period, amplitude, or number of samples). You can, however, enter temporary values to see what a waveform will look like once values such as period, amplitude, and offset, are applied.

Once a wave definition is complete in the WaveDesigner, MSW imports it and applies real timing numbers (Fs and N). The amplitude and offset are not applied at this time, however, as that happens later in the Signal test element.

To summarize, creation of a signal using the MSW consists of three steps:

1. [The WaveDesigner provides the shape of the signal.](#)
2. [The MSW applies the timing.](#)
3. [The test elements assign specific amplitude and offset values.](#)

Note that capture instruments do not use wave definitions since they are capturing samples, not building a signal. Therefore, the WaveDesigner is not used when programming a capture

[instrument.](#)

[See About WaveDesigner](#) for more information.

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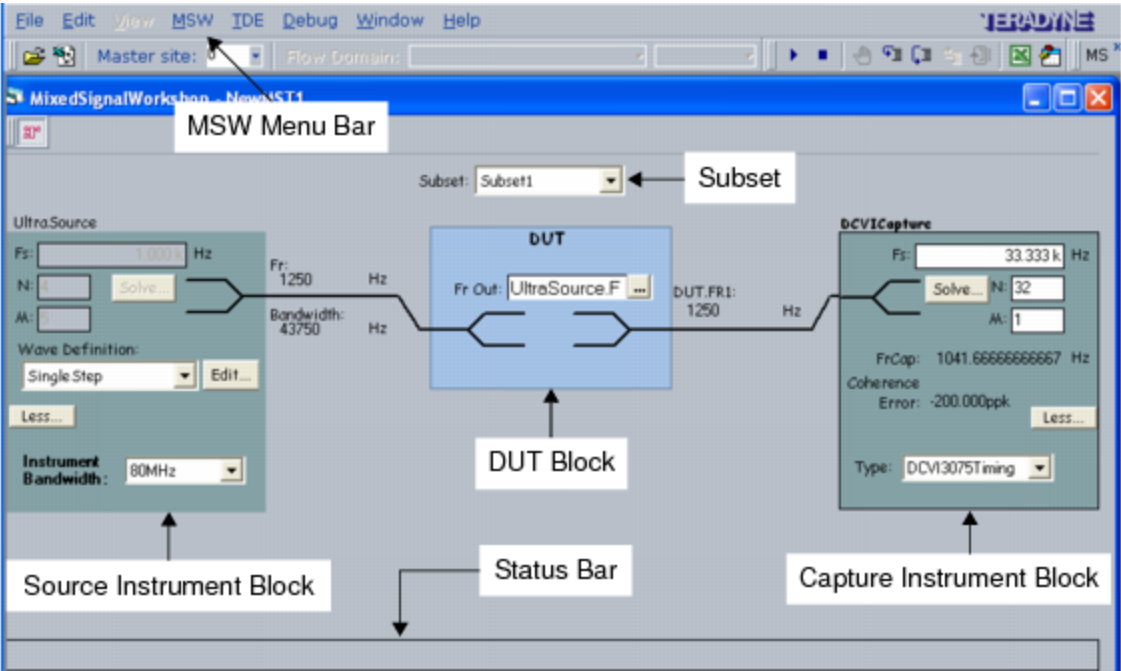
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Parts of the MSW Window

Parts of the MSW Window

The Mixed Signal Workshop window is the display you use to build a simulation of the relationship between the instruments and the DUT used in the test. The window includes the following functional parts:

- MSW Menu Bar
- Source Instrument Block
- Capture Instrument Block
- DUT Block
- Subsets
- Status Bar



Mixed Signal Workshop Window

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Parts of the MSW Window : MSW Menu Bar

MSW Menu Bar

The MSW menu bar is the interface between a Mixed Signal Timing sheet and the MSW. The MSW menu bar has three MSW-specific menus: File, Edit, and MSW. Commands associated with these menus are defined in the following topics:

- File Menu Commands
- Edit Menu Commands
- MSW Menu Commands

	
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Parts of the MSW Window : [MSW Menu Bar](#) : File Menu Commands

File Menu Commands

[File](#) menu commands allow you to open a new or existing MSW timing set. You can also use these commands to save or close timing sets that are active in the MSW.

New	 (Ctrl+N) Creates a new timing set. See Opening a New or Existing Timing Set .
Open	 (Ctrl+O) Opens an existing timing set for creating a new timing set. See Opening a New or Existing Timing Set .
Close MSW	 (Ctrl+W) Closes the active timing set and display, but maintains the active Test Development Environment (TDE). See Closing the MSW .
Save To Excel	 Updates the Mixed Signal Timing sheet with the set information from the active MSW. If a Mixed Signal Timing sheet does not already exist in the workbook, a sheet is created. See Save a Timing Set to the Workbook .
Save To Excel As	Makes a copy of a timing set and saves it to an IG-XL workbook using a different user-specified name. If a Mixed Signal Timing sheet does not already exist in the workbook, a sheet is created. See Save a Timing Set to the Workbook .

Exit TDE

Closes the active MSW set and display, as well as the Test Development Environment (TDE). See [Exiting TDE from the MSW Display](#).

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Parts of the MSW Window : [MSW Menu Bar](#) : Edit Menu Commands

Edit Menu Commands

Creating subsets of a timing set allows you to work around the restriction of only being allowed one timing set per test instance. Edit menu commands allow you to manage subsets of a timing set.

New Subset	Creates a new subset of the current timing set.
Delete Subset	Deletes the active subset of the current timing set.
Rename Subset	Overwrites the current name of the subset with a newly specified name.

For more information, see [Using MSW Subsets](#).

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Parts of the MSW Window : [MSW Menu Bar](#) : MSW Menu Commands

MSW Menu Commands

Using [commands in the](#) MSW menu, you can build a model for determining timing solutions for a test. Use the following commands to add and remove source and capture instruments from a simulation model.

Subset Manager	This feature is not yet implemented.
Add Source	Adds an initial, or an additional, source instrument to the MSW display. To differentiate between multiple instruments of the same type within the active subset, assign that instrument a unique extension identifier called a Resource ID. See Source Instrument Block .
Add Capture	Adds an initial, or an additional, capture instrument block to the MSW display. To differentiate between multiple instruments of the same type within the active subset, assign that instrument a unique extension identifier called a Resource ID. See Capture Instrument Block .
Delete Instrument	Removes a source or a capture instrument block from the active MSW environment.

	
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Parts of the MSW Window : Source Instrument Block

Source Instrument Block

A source instrument block represents a generic source instrument. It shows a signal’s timing and coherence information for a simulated source instrument. Certain instrument types may also have an expanded source block, which displays instrument-specific options for the currently selected instrument.

While the MSW is mainly used to plan and create timing sets, you can use values specified in a source instrument block to program hardware if the Load From Mixed Signal Timing Set button is selected on a Signal test element. For example, when this button is selected on a BBACSourceSignal test element in a test procedure, IG-XL uses data specified in a MSW timing set and stored on a Mixed Signal Timing sheet to program the BBAC Source instrument.

The following topics are available:

- | | |
|---|--|
| • | Signal Timing Information |
| • | Solve Button |
| • | Output Values |
| • | Expanded Source Instrument Block Options |

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Parts of the MSW Window : [Source Instrument Block](#) : [Signal Timing Information](#)

Signal Timing Information

Signal timing information represents parameters that specify the shape, frequency, and resolution of a signal to be created and output by the source instrument. You can [hardcode the sample frequency \(Fs\)](#), [number of samples \(N\)](#), and [number of cycles \(M\)](#) parameters by typing directly in the field, or you can use the Solver to calculate and automatically add these values. [Information about a signal’s shape is based on a wave definition selected from the](#) Wave Definition field menu. To edit a wave definition:

1.

[Select a definition name and click the](#) Edit button to start the WaveDesigner.
2.

[Edit the wave definition in the WaveDesigner.](#)
3.

[Save the wave definition and return to the MSW.](#)

The following controls allow you to specify wave definition and timing values for a source signal:

Fs	The sampling frequency which is the rate at which samples are produced by the source instrument.
N	The number of samples contained within M cycles of the signal.
M	The number of cycles represented by the given set of samples.
Wave Definition	A menu of wave definition names created using the WaveDesigner.
Edit	Invokes the WaveDesigner for editing or creating wave definitions.

For more information, see [Using the MSW Solver](#).

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Parts of the MSW Window : [Source Instrument Block](#) : Solve Button

Solve Button

The Solve button starts the MSW Solver tool, which helps you compute a source instrument’s clocking values based on DUT, DSP, and instrument constraints.
[For more information, see MSW Solver Tool.](#)

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Parts of the MSW Window : [Source Instrument Block](#) : Output Values

Output Values

Using the specified signal timing information, the MSW automatically calculates the repeat frequency (Fr) and Bandwidth of the signal output from a source instrument block.

Fr	The repeat frequency for signal. This is the rate at which the signal repeats itself, or is periodic.
Bandwidth	The highest component in a signal. For example, if the signal is a multitone with two tones at 1.5 kHz and 2 kHz, then FR = 500 Hz and Bandwidth = 2 kHz.

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[Parts of the MSW Window](#) : [Source Instrument Block](#) : Expanded Source Instrument Block Options

Expanded Source Instrument Block Options

Clicking the [More](#) button that appears on some source instrument blocks causes the block to expand and display options unique to the instrument. Typically, these options inform the MSW of any instrument-specific restrictions concerning an instrument's timing or sampling limits. The MSW uses this information to provide valid timing solutions that conform to the instrument specifications.

For example, an instrument may have [multiple low-pass filter settings](#). This implies that each filter has a different maximum sample rate that can be used with it. For the MSW to provide valid timing solutions that take this restriction into consideration, it must know the filter setting and its maximum sample rate.

For information on instruments that have optional parameters in their expanded instrument block, [click Related Topics](#).

	
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Parts of the MSW Window : Capture Instrument Block

Capture Instrument Block

A capture instrument block represents a generic capture instrument. It displays a capture’s timing settings and any coherence error. Certain instruments may also have an expanded capture block, which displays instrument-specific options for the currently selected instrument.

While the MSW is mainly used to plan and create timing sets, you can use values specified in a capture instrument block to program hardware by using the VBT method `.ApplyMixedSignalTiming`. For example, using this syntax in a test program for the BBAC Capture instrument causes IG-XL to use data specified in the MSW timing set and stored on a Mixed Signal Timing sheet to program the BBAC Capture instrument.

The following topics are available:

- | | |
|---|---------------------------|
| • | Signal Timing Information |
|---|---------------------------|
- | | |
|---|--------------|
| • | Solve Button |
|---|--------------|
- | | |
|---|---|
| • | Expanded Capture Instrument Block Options |
|---|---|

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Parts of the MSW Window : Capture Instrument Block : Signal Timing Information

Signal Timing Information

Signal timing information represents parameters that specify the shape, frequency, and resolution of a signal to be acquired by a capture instrument. You can [hardcode the sample frequency \(Fs\)](#), number of samples (N), and number of cycles (M) parameters by typing directly in the field, or you can use the Solver to calculate and automatically add these values.

The [capture frequency \(FRCap\)](#) and the Coherence Error fields are read-only values calculated by the MSW based on the Fs, N, and M settings in the capture instrument block.

Fs	The sampling frequency which is the rate at which samples are captured by the capture instrument.
N	The number of samples contained in M cycles of the signal.
M	The number of cycles represented by the given set of samples.
FrCap	The capture frequency of interest. This is a read-only display whose value is based on the formula $FrCap = Fs * M / N$.
Coherence Error	The discrepancy between FrCap and the DUT.FRx output frequency is the coherence error. This is a read-only display whose value is calculated by the MSW Solver. See DUT Block .

For more information, see [Using the MSW Solver](#).

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Parts of the MSW Window : [Capture Instrument Block](#) : Solve Button

Solve Button

The Solve button starts the MSW Solver tool, which computes a capture instrument’s clocking values based on DUT, DSP, and instrument constraints.

[For more information, see MSW Solver Tool.](#)

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Parts of the MSW Window : [Capture Instrument Block](#) : Expanded Capture Instrument Block Options

Expanded Capture Instrument Block Options

Clicking the More button that appears on some capture instrument blocks causes the block to expand and displays options unique to the instrument. Typically, these options inform the MSW of instrument-specific restrictions concerning an instrument’s timing or sampling limits. The MSW uses this information to provide valid timing solutions that conform to the instrument specifications.

For example, an instrument may have multiple low-pass filter settings. This implies that each filter has a different maximum sample rate that can be used with it. For the MSW to provide valid timing solutions that take this restriction into consideration, it must know the filter setting and its maximum sample rate.

Click the Related Topics button below for a list of instruments that have optional parameters in their expanded instrument block.

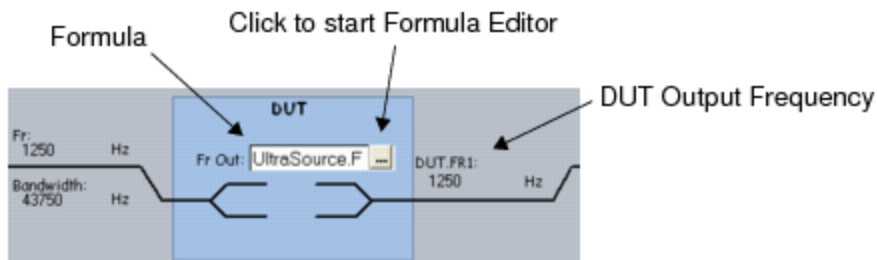
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Parts of the MSW Window : DUT Block

DUT Block

The DUT block models the relationship between the input and output pins on the DUT, which makes it easier to create a coherent relationship between the source and capture frequencies. This block consists of the Formula field (Fr.Out + box), the DUT Output frequency (DUT.FR) value, and an ellipsis button (...) for starting the Formula Editor.



DUT Block

The Formula field is where you build a formula using a source instrument's output frequency as a variable to represent the input frequency of the DUT. This formula can take the form of an arbitrarily complex arithmetic equation. Typically, this formula is defined using variables that describe a source instrument's repeat frequency (Fr). This models what the DUT does to an input signal. For example, if the DUT is a frequency doubler, then the DUT.Out might look like this:

DUT.Out = BBACSrc.FR * 2

In this example, BBACSrc.FR is a variable describing the output repeat frequency of the source instrument used as input to the DUT.

Click the ellipsis button next to the Formula field to invoke the MSW Formula Editor, which helps you to build the DUT output frequency as a formula. For more information, see [Parts of the Formula Editor Display](#).

There is DUT output (DUT.FR) for each capture instrument located on the left side of the DUT block. This read-only value represents the output frequency of the DUT, which is automatically calculated using the formula described in the DUT block's Formula field.

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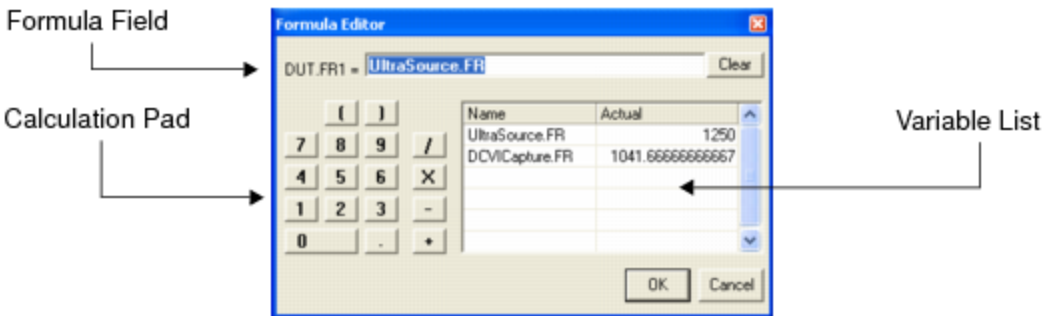
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Parts of the MSW Window : DUT Block : Parts of the Formula Editor Display

Parts of the Formula Editor Display

The Formula Editor allows you to specify certain parameter values as mathematical equations rather than constant values. This allows you to specify frequencies using other frequencies.



Formula Editor

For example, the output frequency of the DUT may equal twice the input frequency of the DUT. This can be represented as a formula:

DUT.FR Out = 2* DUT.FR In

Buttons and fields on the Formula Editor display can be grouped into the following functional parts:

- Formula Field
- Calculation Pad
- Variable List

Formula Field

The Formula field is where you set the formula for the parameter (frequency) in relation to another parameter. This formula can take the form of an arbitrarily complex arithmetic equation.

A formula can use the following operations: addition, subtraction, multiplication, and division, as well as parentheses (to specify the order of operation). A list of variables is created by the MSW, which you can use rather than inputting constants. You are limited to this list in your use of variables for the formula.

Specify a formula in the Formula field either by using the calculation pad and variable list or by typing it in directly.

The Clear button to the right of the Formula field deletes the contents of the Formula field.

Calculation Pad

The Calculation Pad is similar to a basic calculator. Click the Calculation Pad buttons to specify numbers, mathematical operators, or parentheses in the Formula field.

Variable List

A list of variables is created by the MSW based on the instrument resources in the current MSW timing set. You can use these variables rather than inputting the values they represent.

The display shows the name of the variable and its current actual value. The list represents the input and output frequencies of the instruments and the DUT. Each variable's name describes the resource type, resource ID, and the parameter (frequency).

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Parts of the MSW Window : Subsets

Subsets

A timing set defines the timing solution for all the instruments in a particular test instance. Each test instance can have only one timing set. If you want a given instrument to source or capture multiple times with different timing solutions within the same test instance, you must create subsets of that timing set. Each subset describes a unique timing solution for one or more instruments.

Subsets are related to the [VBT method](#) .ApplyMixedSignalTiming. When used in a test program, this method applies the currently active subset of the currently active set to the instrument. The currently active set is defined in a Test Instances sheet. Which subset of the set becomes active is defined by using the VBT syntax TheExec.MST.Subsets. Every set has at least one subset and the first subset is the active subset by default.

For example, to perform two captures, each with a different capture frequency (FR), within the same test instance, while using the same capture instrument, you must describe a capture solution for each FR value in two different subsets. You would then set the first subset active, use .ApplyMixedSignalTiming to apply the data to the capture instrument, perform the capture, and set the second subset active, use .ApplyMixedSignalTiming to apply the data to the capture instrument, and perform the second capture.

When you view the MSW, it shows a complete subset. Change the subset to view or edit by choosing from the Subset drop-down menu. For more information, see [Using MSW Subsets](#).



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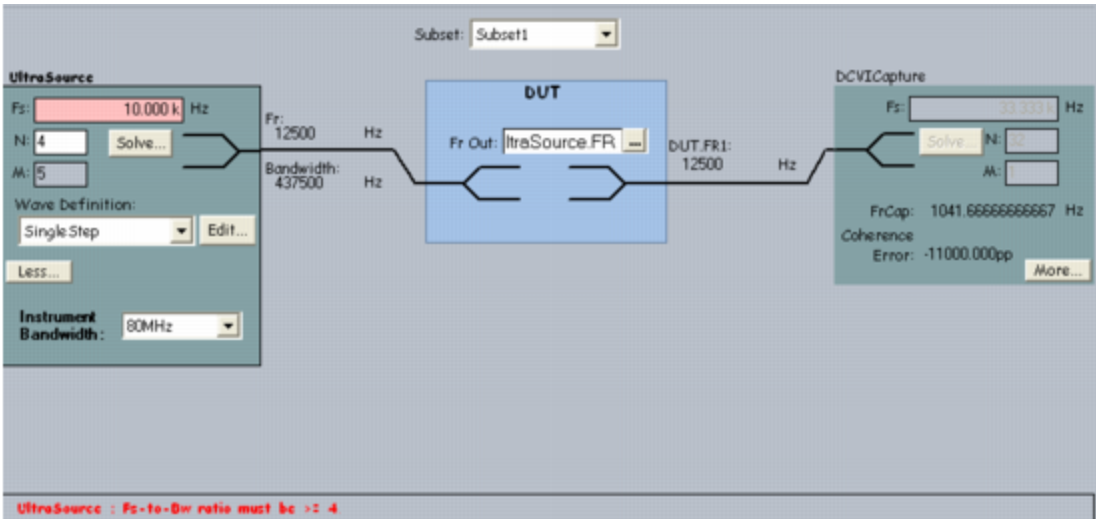
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Parts of the MSW Window : Status Bar

Status Bar

If you perform an invalid operation, or enter an invalid value in the MSW, an error message appears in the lower status bar. The error message explains why the error occurred. For example, the figure shows that an error message appeared because the bandwidth specified was invalid for that type of instrument.



Error Message in the Status Bar

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MSW Solver Tool

MSW Solver Tool

While the Mixed Signal Workshop (MSW) creates coherent timing solutions between instruments, the MSW Solver creates timing solutions for the instruments used in the MSW timing set.

Creating a timing solution for an instrument means that the instrument’s timing values respect the timing equation:

(1) $\frac{FS}{N} = \frac{FR}{M}$

where:

FS	Sample frequency of the instrument (samples/second).
FR	Fundamental frequency of the input or captured signal (Hz).
N	Number of samples in a sourced or captured segment.
M	Number of cycles of the signal generated or captured.

The MSW Solver helps you to calculate a list of all possible timing solutions that satisfy the frequency targets values, constraints, and instrument restrictions.

The following topics are included in this section:

- Starting the MSW Solver
- Parts of the MSW Solver Window
- Using the MSW Solver

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MSW Solver Tool : Starting the MSW Solver

Starting the MSW Solver

Open the MSW Solver from within the MSW. To start the MSW Solver, click the Solve button on the Mixed Signal Workshop's Source or Capture instrument blocks.

Ultra Source

Is 1 JJJJ Is

4 Solve...

5

Wave button bar

Single Step Edit...

Auto...

Instrument Bandwidth 20MHz

Click the Solve button to open the MSW Solver

MSW Solver

Solver

Select the item to be computed: ☒ FS ☐ TR ☐ B ☐ M

Sample Rate (FS) Target 1.000E+12 Tolerance +/- %

Repeat Frequency (FR) Target theSource.FT Tolerance +/- %

Sample Rate (M) 32 4800 powered 2 multiple of

Number of Cycles 1000 1000 multiple of

Other Solution Criteria ☐ Min relative error Optimize for ☐ ☐

N	M	FS	FR	FR error	Time	LTP limit	Selected Level
40	-	75,000,000	1,250,000	0.0	1.250 000	3 333	
42	-	77,500,000	1,250,000	0.0	1.250 000	3 333	
44	-	80,000,000	1,250,000	0.0	1.250 000	3 333	
46	-	82,500,000	1,250,000	0.0	1.250 000	3 333	
48	-	85,000,000	1,250,000	0.0	1.250 000	3 333	
50	-	87,500,000	1,250,000	0.0	1.250 000	3 333	

Show Changes ☐ Show all 1979 solutions found 2000 evaluated

OK Cancel

Opening the MSW Solver

If you already entered values in instrument block fields before opening the Solver, these values are displayed in the relevant fields in the Solver.

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MSW Solver Tool : Parts of the MSW Solver Window

Parts of the MSW Solver Window

The MSW Solver window is an application that helps you solve one of the four timing variables based on the other three. As such, the fields and buttons on the Solver display are dynamic and react based on the variable you want to solve for.

Solver

Select the item to be computed:

☒ FS

☐ FR

☐ N

☐ M

=

Sample Rate (FS)

Target: 1,000 Hz

Tolerance: +/- %

Repeat Frequency (FR)

Target: traSource.FR ...

Tolerance: +/- %

Sample Count (N)

32 to 4096

☐ power of 2

☒ multiple of 2

2

Number of Cycles (M)

1 to

☐ odd

☒ multiple of 1

1

Other Solution Criteria

☐ M,N relatively prime

Optimize for:

☐ FS

☒ FR

N	M	FS	FR	IFR error	Fres	UTP (ms)	Rejected becau:
140	1	175,000.000	1,250.000	0.0	1,250.000	0.800	
142	1	177,500.000	1,250.000	0.0	1,250.000	0.800	
144	1	180,000.000	1,250.000	0.0	1,250.000	0.800	
146	1	182,500.000	1,250.000	0.0	1,250.000	0.800	
148	1	185,000.000	1,250.000	0.0	1,250.000	0.800	
150	1	187,500.000	1,250.000	0.0	1,250.000	0.800	

Show Choices

☐ Show all

1979 solutions found
2033 evaluated

OK

Cancel

MSW Solver Window

The display includes:

- The timing equation, which you use to select the variable you want to solve for.
- Regions to specify timing and sampling values from which the Solver calculates the unknown variable's value.
- The Other Solution Criteria region, which allows you to impose restrictions on the Solver's recommended results.
- A results table, where the Solver lists all the possible, or just the valid, solutions based on the information provided.

The following topics describe using the Solver's fields and buttons:

- Timing Equation
- Sample Rate (FS)
- Repeat Frequency (FR)
- Sample Count (N)
- Number of Cycles (M)
- Other Solution Criteria
- Results Table

Solver Error Checking of Entered Values

As you enter values into the Solver text boxes, the tool does some limited error checking. However, the valid range for each field varies for each UltraFLEX instrument, and the Solver may not accurately validate values you enter in some cases.

This may allow you to enter values into Solver fields that are outside the range tolerated by a particular instrument. However, solutions generated by the Solver for an instrument will all be within the limits tolerated by that instrument.

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MSW Solver Tool : Parts of the MSW Solver Window : Timing Equation

Timing Equation

The MSW Solver helps you achieve a timing solution by calculating valid values that satisfy the timing equation.

(2) $\frac{FS}{N} = \frac{FR}{M}$

where:

FS	Sample frequency of the instrument (samples/second).
FR	Fundamental frequency of the input or captured signal (Hz).
N	Number of samples in a sourced or captured segment.
M	Number of cycles of the signal generated or captured.

An interactive version of the timing equation appears at the top of the Solver window. By selecting one of the four variables in the equation, you identify the unknown variable for which the Solver will calculate a valid value based on the other three variables. Note that when you select a variable, the fields related to that variable become inactive, while the other three variable regions become active. This reinforces the idea that you must specify values for the three variables, and that the Solver uses this information to generate valid values for the desired variable.

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MSW Solver Tool : [Parts of the MSW Solver Window](#) : Sample Rate (FS)

Sample Rate (FS)

The Sample Rate (FS) section of the MSW Solver display contains the following fields:

- Target
- Tolerance +/-
- Undersample (Capture only)

Target

The Target field allows you to specify a desired value for the sample frequency of the instrument. The target sample rate value is measured as the number of samples per second that the instrument is sourcing or capturing. This field is always active on the Solver display, unless you select FS in the Timing Equation.

Tolerance

The Tolerance field allows the Solver to restrict valid solutions to those that have a sample rate (FS) within a percentage of the target FS. The value you enter in the Tolerance field represents the percentage the solutions can deviate from the target sample rate. For example, if the target FS = 200 MHz and the tolerance is 10, then the Solver will show all valid solutions whose FS value is between 180 and 220 MHz.

The Tolerance field is only active on the Solver display when Optimize for FR is selected, and you are solving for M or N.

If you do not specify a tolerance value, the Solver shows only solutions with the best sample rate (FS) accuracy. This accuracy may not be the desired FS, but it will be the closest possible value based on the restrictions imposed on the solution.

For more information, see [When to Use Tolerance](#).

Undersample

The Undersample field appears only when you open the Solver from a capture instrument block. When selected, it allows the Solver to use an undersampling capture technique to determine the effective sampling rate. Use undersampling when the desired sample rate is faster than the instrument can go.

When you select Undersample, the Solver displays the following equation which is used to determine the effective sample rate:

$$FS\ effective = FS\ real * USR$$

where:

FS effective	The effective sampling rate of the capture.
FS real	The actual sampling rate of the instrument hardware.
USR	The Undersampling Ratio (USR) is a positive integer ratio between FS effective and FS Real.

Selecting Undersample allows the Solver to use a USR value that is greater than 1 to achieve an effective sample rate that is as close to the target FS as possible. For example, if the target FS is 1.4 MHz, but the instrument has a maximum sample rate of 1 MHz, the Solver will choose solutions that have FS effective equal 1.4 MHz, FS Real equal 700 kHz, and USR equal 2. Without undersampling the Fs would equal 1 MHz, its maximal sample rate.

You can specify the USR value, which forces the Solver to use that USR for all its solutions. If you do not specify a USR value, the Solver uses the smallest USR that achieves the Target FS.

Note that it is the FS Real value that is displayed in the MSW display's capture instrument block when the solution is applied.

For more information, see [When to Use Undersampling](#).

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MSW Solver Tool : [Parts of the MSW Solver Window](#) : Repeat Frequency (FR)

Repeat Frequency (FR)

The Repeat Frequency (FR) section of the MSW Solver display contains two fields:

- | | |
|---|-------------------------------|
| • | Target |
| • | Tolerance +/- |

Target

The Target field allows you to specify a value for the fundamental frequency of the input or captured signal. The target repeat frequency value is measured as the number of times the signal repeats itself and its measured in Hertz. This field is always active on the Solver display, unless you select FR in the Timing Equation.

[Specify the target value of FR as a value or as an equation. To specify the target FR as an equation, click the ellipsis button \(...\) next to the Target field and create the variable using the Formula Editor](#)
[For more information, see \[Parts of the Formula Editor Display\]\(#\).](#)

Tolerance

The Tolerance field allows the Solver to restrict valid solutions to those that have a repeat frequency (FR) within a percentage of the target FR. The value you enter in the Tolerance field represents the percentage for which the solutions can deviate from the target repeat frequency. For example, if the target FR = 200 MHz and the tolerance is 10, then the Solver will show all valid solutions whose FR value is between 180 and 220 MHz.

The Tolerance field is only active on the Solver display when Optimize for FS is selected, and you are solving for M or N.

[If you do not specify a tolerance value, the Solver shows only the solutions with the best repeat frequency \(FR\) accuracy. This accuracy may not be the desired FR, but it will be the closest possible](#)

value based on the restrictions imposed on the solution.
For more information, see [When to Use Tolerance](#).

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MSW Solver Tool : [Parts of the MSW Solver Window](#) : Sample Count (N)

Sample Count (N)

The Sample Count (N) section of the MSW Solver display contains the following fields:

- [Sample Count Range](#)
- [power of 2](#)
- [multiple of <x>](#)

Sample Count Range

The Sample Count Range field restricts the sample count (N) values to those that fall within the range specified. If you specify only the minimum value (first box), then only that N value is considered valid and the other conditions (power of 2 and multiple of <x>) are ignored. This field is always active on the Solver display, unless N is selected in the Timing Equation.

Power of 2

The power of 2 field restricts the value of N to those that are a power of 2. For example, if the specified range is 10 to 100, and the power of 2 field is selected, the Solver only generates solutions whose N values are powers of 2 between 10 and 100 (16, 32, 64...).

Multiple of <x>

The multiple of <x> field restricts the sample count (N) value to those that are a multiple of x, where x is a positive integer. For example, if the specified range is 10 to 100, and the multiple of <x> field is set to 3, the Solver only generates solutions whose N values are multiples of 3 between 10 and 100 (12, 15, 18...).

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MSW Solver Tool : [Parts of the MSW Solver Window](#) : Number of Cycles (M)

Number of Cycles (M)

The Number of Cycles (M) section of the MSW Solver display contains the following fields:

- Number of Cycles Range
- odd
- multiple of <x>

Number of Cycles Range

The Number of Cycles Range field restricts the number of cycles (M) values to those that fall within the range specified. If you specify only the minimum value (first box), then only that M value is considered valid and the other conditions (odd and multiple of <x>) are ignored.

This field is always active on the Solver display, unless M is selected in the Timing Equation.

Odd

The odd field restricts the value of M to those that are odd. For example, if the specified range is 10 to 100, and the odd field is selected, the Solver only generates solutions whose M values are odd between 10 and 100 (11, 13, 15...).

Multiple of <x>

The multiple of <x> field restricts the number of cycles (M) value to those that are a multiple of x, where x is a positive integer. For example, if the specified range is 10 to 100, and the multiple of <x> field is set to 3, the Solver only generates solutions whose M values are multiples of 3 between 10 and 100 (12, 15, 18...).

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MSW Solver Tool : Parts of the MSW Solver Window : Other Solution Criteria

Other Solution Criteria

The Other Solution Criteria section of the MSW Solver display contains the following fields:

- M,N relatively prime
- Optimize for FS/FR
- Minimum Capture Bandwidth (Capture only)

M, N Relatively Prime

When you select the M, N relatively prime field, the Solver restricts valid solutions to those that have M and N values that are relatively prime to one another. In other words, it ensures that the greatest common denominator between M and N is 1.

Optimize for FS, FR

When solving for M or N, you are trying to achieve both the target FR and the target FS. Often, you cannot achieve both these values without error. The Optimize field finds the best FS while remaining within the Tolerance bounds of FR, and vice versa.

Selecting Optimize for FS assigns priority to find the best FS, while remaining within the bounds of the specified tolerance. Selecting Optimize for FR assigns priority to find the best FR, while remaining within the bounds of the specified tolerance.

This field also activates the Tolerance field in the FS and FR regions of the display.

Minimum Capture Bandwidth

The Minimum Capture Bandwidth field appears only when the MSW Solver is opened from a capture instrument block. This field allows you to specify the frequency that the solution must be able to capture according to the Nyquist criteria. This places an extra restriction on what is

considered a valid FS for the solution. If you do not specify a minimum capture bandwidth, no extra restrictions are applied to the solutions.

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MSW Solver Tool : Parts of the MSW Solver Window : Results Table

Results Table

The Results Table activates and lists solutions generated by the Solver based on data provided in the FS, FR, M, N, and Other Solution Criteria fields of the display.

Click a column heading to sort the table based on that column.

N	M	FS	FR	FR error	Fres	UTP (ms)	Rejected because
140	1	175,000,000	1,250,000	0.0	1,250,000	0.800	
142	1	177,500,000	1,250,000	0.0	1,250,000	0.800	
144	1	180,000,000	1,250,000	0.0	1,250,000	0.800	
146	1	182,500,000	1,250,000	0.0	1,250,000	0.800	
148	1	185,000,000	1,250,000	0.0	1,250,000	0.800	
150	1	187,500,000	1,250,000	0.0	1,250,000	0.800	

Show Choices

Show all

1979 solutions found
2033 evaluated

OK

Cancel

Select Show All to view all generated solutions, even those that fail because of various constraints.

Displays the number of solutions generated.

MSW Solver Results Table

Once you complete the Solver fields, click the Show Choices button. The Solver evaluates solutions based on the specified criteria and lists the solutions in the Results Table.

The table lists the features of the Results Table.

Results Table Features

Column/Button	Description
N	The number of samples used to represent the given signal.
M	The number of cycles represented by the given set of samples.
FS	The sampling frequency, which is the rate at which samples are produced by the source instrument.
FS error	The discrepancy between the target FS and the solution's FS.

FS effective	The effective sampling rate of the capture. (Capture only.)
FS real	The actual sampling rate of the instrument hardware. (Capture only.)
USR	The Undersampling Ratio (USR) is a positive integer ratio between FS effective and FS Real. (Capture only.)
FR	The repeat frequency for signal. This is the rate at which the signal repeats itself, or is periodic.
FR error	The discrepancy between the target FR and the solution's FR.
Fres	Frequency Resolution (FRes) $FRes = FS/N$
UTP (ms)	Unit Test Period is a measure of how long a signal will take to be sourced or captured. $UTP = N/FS$.
Rejected because	Description of why a solution was deemed invalid. Rejected solutions only appear when Show all is selected.
Show Choices	Activates the Solver to start evaluating solutions based on the data specified in the FS, FR, M, N, and Other Solution Criteria fields of the display.
Stop	When you click Show Choices, the Solver starts to evaluate solutions, and the Show Choices button becomes the Stop button. Click Stop to stop the Solver from evaluating any more solutions.
Show all	When selected, all solutions evaluated by the Solver (valid and invalid) are listed in the Results Table. Rejected solutions will include a description of why they were invalid in the Rejected because column.
<x> solutions found	The number of valid solutions that were evaluated by the Solver.
<x> evaluated	The total number of solutions the Solver generated (valid and invalid).

To sort the table by columns, click a column's heading. After evaluating which solution best meets your needs, select the solution's row in the table and click OK. The solution values are applied to the instrument block in the Mixed Signal Workshop window.

☑ Note:

Double-clicking the solution row automatically closes the Solver window and imports the solution data into the MSW instrument block.

For more information, see [Choosing a Timing Solution](#).

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[MSW Solver Tool](#) : Using the MSW Solver

Using the MSW Solver

The Solver is flexible, allowing you to use it in a number of ways to find the best timing solution for your application. The following topics provide a general use model for the tool. Guidelines and recommendations for what variables to solve for, what constraints to apply, and what solutions to choose are described in the following topics:

- | | |
|---|---|
| • | Using the Solver: General Steps |
| • | When to Solve for FS, FR, M, or N |
| • | Solving for M or N |
| • | Choosing a Timing Solution |
| • | When to Use Undersampling |
| • | When to Use Tolerance |

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MSW Solver Tool : Using the MSW Solver : Using the Solver: General Steps

Using the Solver: General Steps

The following is the general procedure for using the Solver to build an instrument’s timing solution:

1.

Designate one of the four variables as the dependent variable by selecting the radio button that represents the variable to be computed. That is, the one to be calculated from the other three.
2.

Specify values for the other three variables. Set constraints, targets, ranges, and priorities to limit the list of choices to a manageable number.

–

Constrain the possible values of M and N by specifying a value or a range or by applying a constraining condition.

–

Set targets, tolerance, and priorities (Optimize for) for FS and FR.

3.

Click the Show Choices button to have the Solver enumerate solutions, adding those that satisfy the constraints to the Results Table.
- The Solver enumerates combinations of the three independent variables and, for each combination, computes the value of the selected dependent variable.
- Any solution that does not meet both of the following criteria is discarded:
- FS and N must be precisely realizable by the instrument. If necessary, FS and, therefore, FR are modified slightly to conform to the instrument’s resolution.
- file:///C:/Users/cpatle2/AppData/Local/Temp/chm_extract_9x9mffc4/combined.html

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- The instrument must achieve the required bandwidth for the values and options chosen.
- The timing solution must satisfy the Nyquist criteria.

The number of solutions found and evaluated are displayed next to the Show Choices/Stop button. Valid solutions are listed in a tabular format along with various calculated quantities and metrics such as Fres, UTP, and FS/FR errors. If the Solver does not produce any valid solutions, an error message appears at the bottom of the Solver window. Correct the error and click the Show Choices button again.

4. If the Solver is generating more solutions than you need, click the Stop button to interrupt the Solver and work from the list of already generated solutions.
5. To better evaluate solutions, sort the table on the various columns by clicking a column's heading. This allows you to prioritize and evaluate solutions based on what variables are most important to your application. For more information, see Choosing a Timing Solution.
6. Once you identify the solution to be used, select the solution's row and click OK. The solution values are transferred to the MSW tool.

			
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MSW Solver Tool : Using the MSW Solver : When to Solve for FS, FR, M, or N

When to Solve for FS, FR, M, or N

Which variable do you solve for? In general, you solve for the variable that has the least impact on your testing requirements, or the variable that you have the most flexibility to change. You also solve for FS for analog source or capture instruments and either N or M for digital source or capture instruments.

The reason you solve for FS for analog instruments is that the value of the FS number in and of itself means nothing to you, and there is nothing forcing it to be one value or another. The FR value, however, is always important, and the N and M values are important to all capture instruments since they affect the DSP analysis of data.

The reason you solve for N or M for digital instruments is that FS is usually forced to be one number by the pattern. This is because digital instruments derive their FS directly from the speed the pattern is running at, and the pattern must run at a certain speed as dictated by the DUT. The FR is again always important. The N and M values are also important both for digital source and capture, but you have the most flexibility to change those values, which you do not for FR and FS.

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MSW Solver Tool : [Using the MSW Solver](#) : Solving for M or N

Solving for M or N

When solving for FS or FR, two of the three independent variables (M and N) are discrete, which severely limits the number of solutions to be evaluated. When solving for M or N, however, two of the three (FS and FR) are continuous, which theoretically requires a nearly infinite number of evaluations.

The Solver deals with this problem by requiring you to constrain both FS and FR in one of two ways:

- [Using Optimize for FS/FR](#), you can set a target value for each of the frequencies, with one of the two identified as the priority frequency. The Solver then tries to achieve the priority frequency as precisely as possible, while coming as close as possible to the other target (see [Optimize for FS, FR](#)).
- [Using a Tolerance value](#), you can set a target value for one frequency and select a range for the other. Tolerance filters out solutions that have more error from the target than the tolerance value allows. The targeted frequency is automatically set to the priority frequency (see [Tolerance](#)).

In either case, the priority frequency is achieved within the resolution of the instrument (typically parts-per-billion or better). The nonpriority frequency may diverge from its target by parts-per-hundred due to the coarse granularity of M and N.

	
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MSW Solver Tool : Using the MSW Solver : Choosing a Timing Solution

Choosing a Timing Solution

Choosing a timing solution is a matter of evaluating your timing needs and choosing a solution that optimizes your testing requirements. The Solver is useful here because it is flexible enough to allow you to choose solutions from a variety of different priorities. In general, to identify which solution to use:

- | | |
|----|---|
| 1. | Determine what tolerance you can accept in the four variables (FS, FR, N, and M). |
| 2. | Sort the solutions in the Solver’s Results Table based on these tolerances. |
| 3. | Identify the solution that has the least error in all the four variables. Often, you are really only looking for error in FS and FR, or even just FR. |

In general, choosing a timing solution is a trade-off between test time and accurate repeatable results. For example, a large number of cycles (M) provides more repeatable results but also longer test times. Which is more important to you? Solutions generated by the Solver provide you with data to compare, and you can make an informed decision based on how you weigh your timing priorities: repeatable results versus test time.

	
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MSW Solver Tool : Using the MSW Solver : When to Use Undersampling

When to Use Undersampling

When the Solver is opened from a capture instrument block, the Sample Rate (FS) region of the display includes the Undersample field (see [Undersample](#)). When selected, it allows the Solver to use an undersampling capture technique to determine the effective sampling rate.

The Nyquist criterion requires that the sampling rate must be at least twice the frequency of the highest frequency component in the signal to be sampled. When the sampling rate is set below this criterion (less than two samples/cycle), it is called undersampling, and under these conditions, aliasing occurs.

Sometimes you want to capture a fast signal with a high repeat frequency (FR). The FR may be so high that the instrument cannot supply a sample rate (FS) fast enough to meet Nyquist. Therefore, use undersampling when the desired sample rate is faster than the instrument can go.

For example, a capture instrument can capture at a FS of 20 MHz, but you want to capture 20 samples per cycle of a 40 MHz signal. That is, FR = 40 MHz, and $N \geq 20$. If you want a $M = 1$, the resulting FS is 800 MHz.

$$(3) \quad FS = N \times \frac{FR}{M} \rightarrow FS = 20 \times \frac{40 \text{ MHz}}{1} \rightarrow FS = 800 \text{ MHz}$$

Since the FS (800 MHz) is much faster than the 20 MHz the instrument is capable of, you must undersample. Using an undersampling ratio (USR) of 40, you can run the instrument at its fastest rate (20 MHz), and it will look as though you're running it at 800 MHz. You can only use this technique if you have a stable, repeatable signal, and your capture instrument has the bandwidth to receive the desired signal.

$$FS \text{ Effective} = FS \text{ Real} \times USR \rightarrow 800 = 20 \times 40$$

$$FS = 20 \times \frac{40 \text{ MHz}}{M}$$

$$FS \times M = 800 \text{ MHz}$$

$$(4) \quad \text{Select } FS = 20 \text{ MHz}$$

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MSW Solver Tool : Using the MSW Solver : When to Use Tolerance

When to Use Tolerance

Both the Sample Rate (FS) and the Repeat Frequency (FR) regions of the Solver display include the Tolerance field (see [Tolerance](#) and [Tolerance](#)). This field allows the Solver to restrict valid solutions to those that have an FS or FR within a percentage of the target.

As described in [Solving for M or N](#), the Solver may not be able to achieve both the target value for FS and for FR. Therefore, specifying a tolerance for these variables provides the Solver with some leeway by deviating from the target as much as is specified by the tolerance value, while still providing an FS or FR value as close to the target value as possible.

For example, you want your timing solution to have FS = 100.0 MHz and FR = 5.02 MHz. However, you may not be able to achieve both of them exactly since both N and M are integer numbers in the timing equation ($FS/N = FR/M$). For the equation to be true, either FS or FR (which are real numbers) must be rounded. To round, you must decide which one of the two has priority, or in the Solver, which should be optimized. (See [Optimize for FS, FR](#).) Then you must set the nonpriority variable's tolerance to determine by how much you are willing to let the nonpriority variable deviate from its target value.

If you decide to optimize for FS and have the Solver generate solutions, you may get the desired FS = 100.0 MHz and FR = 5.1 MHz because of the N and M restrictions you specified. The generated FR value may be too far from the desired 5.02 MHz to be useful. Therefore, you must adjust your N and M restrictions to generate more solutions, or you will never achieve your FR and FS within the accuracy you desire.

The tolerance is a measure as a percentage or target of how far the suggested value can be from the target and still meet your requirements. By specifying a tolerance value, you filter out all solutions that are not close enough. This may result in generating no solutions, at which point you may need to loosen some of your restrictions to get a good solution.

The default state of the Tolerance field is blank. However, this does not mean 0% tolerance. Rather, it passes the following criteria to the Solver:

- Achieve the optimized value (either FR or FS) as close as possible for a given N and M value.
- Calculate only the best other value (either FS or FR).
- Do not list all possible values, only the best one. That best one may have a tolerance error of 0% or 5000%. Whatever it is, it is the best you can do given your restrictions.

If you specify a tolerance, the Solver does not list solutions that exceed that tolerance, even if that means generating no solutions. An error message may appear and suggest that you loosen your restrictions.

		
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Using the MSW Tool

Using the MSW Tool

MSW is an environment where you build a simulation of the relationship between a source instrument, a capture instrument, and the DUT. Using an existing wave definition, you specify clocking requirements for the instruments as relationships and constraints in the MSW Solver. Based on the specified relationships and constraints, the Solver calculates and displays possible clock frequencies that the instruments can be programmed to and still achieve a clocking scenario with zero coherence error. The MSW also allows you to tighten the frequency accuracy requirements, even at the cost of coherence error.

The following topics describe how to operate MSW:

- [Starting the MSW](#)
- [Opening a New or Existing Timing Set](#)
- [Closing the MSW](#)
- [Save a Timing Set to the Workbook](#)
- [Exiting TDE from the MSW Display](#)
- [Adding and Removing Instrument Blocks](#)
- [Previewing the Sourced Waveform](#)

•	Using MSW Subsets
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Using the MSW Tool : Starting the MSW

Starting the MSW

To launch MSW, on the IG-XL Tools tab, in the Launch group, select Tools>MixedSignalWorkshop. The MSW window appears with only a DUT block visible.

You can also start MSW from a Mixed Signal Timing sheet. Right-click in the Set Name column, and select Edit from the dropdown. MSW displays the first subset of the wave.

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Using the MSW Tool : Opening a New or Existing Timing Set

Opening a New or Existing Timing Set

When the MSW is started from TDE, the initial MSW window only contains a DUT block. You must create a new timing set or open an existing one.

Select [File>New](#) to create a new MSW timing set.

The MSW displays a DUT block, plus the [Subset](#) field. You can now add source and capture instrument blocks (see [Adding a Source Instrument Block](#) and [Adding a Capture Instrument Block](#)).

To open an existing timing set:

- | | |
|----|---|
| 1. | Select File>Open or click the Open button  in the MSW tool bar. |
| 2. | Select a timing set name from the list in the Mixed Signal Timing Set dialog box and click OK . |

The timing set appears in the MSW display.



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Using the MSW Tool : Closing the MSW

Closing the MSW

To close the active timing set and exit the MSW tool, while keeping TDE open:

1.

Select File>Close MSW or click the Close button  in the upper-right corner of the MSW tool bar.
2.

If the most recent changes to the timing set have not been saved, the Mixed Signal Workshop dialog box appears and asks if you want to save the active timing set to the Mixed Signal Timing sheet. Choose one of the following:

Yes	Saves the active timing set to the Mixed Signal Timing sheet before closing the MSW tool.
No	Does not save the active timing set to the Mixed Signal Timing sheet before closing the MSW tool.
Cancel	Ignores the close the MSW tool request and returns to the active timing set in the MSW display.

To close the active timing set without closing the MSW tool, either open an existing timing set or follow the procedure for opening a new timing set (see Opening a New or Existing Timing Set).

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Using the MSW Tool : Save a Timing Set to the Workbook

Save a Timing Set to the Workbook

To save information from the active timing set in the MSW display to a Mixed Signal Timing sheet in the IG-XL workbook:

1. [Select](#) File>Save To Excel or click the Save to Excel button  on the MSW tool bar.

The Save to Workbook As dialog box appears if it is the first time that the timing set has been saved. Otherwise, the timing set is automatically saved to the Mixed Signal Timing sheet using the existing name and you do not have to complete step 2.

2. [Enter a name for the timing set in the lower field of the dialog box and click](#) OK.

The timing set is saved to the Mixed Signal Timing sheet using the specified name.
[Note that MSW save options](#) (Save to Excel and Save to Excel As) are not supported if the Mixed Signal Timing sheet has been removed from the workbook. If you use these options without a Mixed Signal Timing sheet, IG-XL returns an error.
[Save an Existing Timing Set Under a Different Name](#)
[Use the](#) Save To Excel As command to save an updated copy of an existing timing set without overwriting the original timing set.

1. [Select](#) File>Save to Excel As.

The Save to Workbook As dialog box appears.

2. [Type a name for the timing set in the lower field of the dialog box and click](#) OK.

The new timing set is saved to the Mixed Signal Timing sheet using the specified name, and the original timing set remains unchanged.

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Using the MSW Tool : Exiting TDE from the MSW Display

Exiting TDE from the MSW Display

You can close the active timing set and exit both the MSW tool and TDE.

To close the TDE from the MSW tool:

1.

Select File>Exit TDE or click the Close button  in the upper-right corner of the TDE window's title bar.
2.

If the most recent changes to the timing set have not been saved, the Mixed Signal Workshop dialog box appears and asks if you want to save the active timing set to the Mixed Signal Timing sheet. Choose one of the following:

Yes	Saves the active timing set to the Mixed Signal Timing sheet before closing the MSW tool.
No	Does not save the active timing set to the Mixed Signal Timing sheet before closing the MSW tool.
Cancel	Disregards the Exit TDE request and returns to the active timing set in the MSW display.

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Using the MSW Tool : Adding and Removing Instrument Blocks

Adding and Removing Instrument Blocks

Adding a Source Instrument Block

When a new timing set is created, the MSW display only creates a DUT block. You must add source instruments manually to the display.

To add a source instrument to the MSW timing set:

- | | |
|----|--|
| 1. | Select MSW>Add Source. |
| 2. | Select a source instrument from the list in the Select Resource dialog block and click OK. |

An error message appears in the status bar if a source instrument of the same type already exists in the active MSW environment.

- | | |
|---|--|
| – | With the source instrument highlighted in the list, type a name in the Resource ID box and click OK. Note that the Resource ID must begin by a letter and can only contain letters, numbers, and the underscore character. |
|---|--|

The new source block is added to the MSW display. Note that the name of the new source block contains the instrument type and Resource ID.

Adding a Capture Instrument Block

When a new timing set is created, the MSW display only creates a DUT block. You must add capture instruments manually to the display.

To add a capture instrument to the MSW timing set:

- | | |
|----|-------------------------|
| 1. | Select MSW>Add Capture. |
|----|-------------------------|

2.
- Select a capture instrument from the list in the Select Resource dialog block and click OK.

An error message appears in the status bar if a capture instrument of the same type already exists in the active MSW environment.

-
- With the capture instrument highlighted in the list, type a name in the Resource ID box and click OK. Note that the Resource ID must begin by a letter and can only contain letters, numbers, and the underscore character.

The new capture block is added to the MSW display. Note that the name of the new capture block contains the instrument type and Resource ID.

Removing an Instrument Resource Block

To remove an instrument from the active MSW timing set:

1.
- Select the instrument resource block in the display so that it is the active component.
2.
- Select MSW>Delete Instrument.

The instrument block is deleted from the active MSW timing set.
An alternative method for removing an instrument block from the timing set is:

1.
- Select the instrument resource block in the display so that it is the active component.
2.
- Right-click over the instrument block and select Delete Instrument from the menu.

The instrument block is deleted from the active MSW timing set.

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Using the MSW Tool : Previewing the Sourced Waveform

Previewing the Sourced Waveform

Right-click an active source instrument block to bring up a menu to view the signal’s samples in WaveScope. You can either view the ideal samples using the Show Waveform command or the simulated samples using the Show Simulated Waveform command.

Ideal samples come directly from the wave definition, meaning that they perfectly represent the signal as defined by the wave definition. Simulated samples are what the ideal samples look like after having been passed through the hardware. This is a software simulation of hardware behavior. These simulations are detailed but not perfect approximations of hardware behavior.

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Using the MSW Tool : Using MSW Subsets

Using MSW Subsets

Each test can have only have one timing set. However, a timing set can consist of multiple clocking solutions, each of which is accessed by layering the timing sets into subsets (see [Subsets](#)). Subsets offer different solutions within the same timing set. This is important for cases when one or more instrument clocks must be reprogrammed to different frequencies during a single test.

When building timing solutions with MSW, you can:

- [Add a new timing subset to the timing set. See Adding a New Timing Subset.](#)
- [View another subset within the timing set by switching subsets. See Switching From One Subset to Another Within MSW.](#)
- [Rename a timing subset. See Renaming a Timing Subset.](#)
- [Delete a timing subset from the timing set. See Deleting a Timing Subset.](#)

By default the first subset in a timing set is the active subset for the test. To assign another subset to be the active timing solution in a test use a VBT language statement to specify the desired subset. See [Setting a Subset as the Current Timing Subset](#).

Adding a New Timing Subset

To add a new subset, select [Edit>New Subset](#). The subset name in the Subset field then increments to the next default name (Subset2, Subset3).

Switching From One Subset to Another Within MSW

All subsets for the active timing set are listed in the Subset field's drop-down menu. Switch from one subset to another by selecting a subset name from the Subset field's menu.

Renaming a Timing Subset

To rename a subset:

1.

Select Edit>Rename Subset.

The Name Subset dialog box appears with the active subset name in the test field.

2.

Select and type over the current name. Click OK.

Be sure to assign the subset a unique name that is not already used in the timing set.

Deleting a Timing Subset

To delete a subset, select Edit>Delete Subset. The current subset is removed from the Subset field, and MSW restores the previous subset in the Subset menu as the current subset.

Setting a Subset as the Current Timing Subset

By default, the first subset in a timing set is the active subset for the test. To assign another subset to be the active timing solution in a test, use the VBT language statement TheExec.MST.Subsets to specify the desired subset.

See MST (Mixed Signal Timing) in the VBT Syntax Reference for information about the VBT language statements that control the setting of timing subsets.

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